
Question	Marking Guidance	Mark	Comments
03.1	(A measure of) the number of (different) species in a community;	1	For 'community' accept 'habitat/ecosystem/one area/environment' Reject 'in a population'.

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03.2	Yes, natural best, because 1. Peak of (mean) bee numbers in natural habitat is highest; 2. The (mean) number of bees was higher in the natural habitat until day 200; 3. (Mean) species richness in natural habitat higher at all times; No, natural not best, because 4. Lowest (mean) number of bees after day 220; Yes, town worst, because 5. Peak of species richness higher in <u>both</u> natural and farmland OR Species richness lowest in town from day 125; No, town not worst, because 6. (Mean) species richness is lower in farmland until day 125; 7. Similar (mean) number of bees to farmland; OR (Mean) number of bees lower in farmland until day 140; General, no, because 8. Index of diversity of bees not measured OR The number of bees of each species is not known;	4 max	1. For accept description for 'peak'. 2. For 'day 200' accept any day between 190 and 210. 2. For 'until day 200' accept 'for 200 days'. 4. For 'day 220' accept any day between 210 and 230. 5 and 6. For 'day 125' accept any day between 115 and 135. 6. For 'until day 125' accept 'for 125 days'. 7. For 'day 140' accept any day between day 130 and 150. 7. For 'until day 140' accept 'for 140 days'

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03.3	1. Must not harm the bees OR Must allow the bee to be released unchanged; 2. Must allow close examination OR Use a key (to identify the species);	2	2. Accept method that allows close examination 2. Ignore references to DNA sequencing 2. Accept 'use photographs/specimens (to identify species)'
03.4	1. Collect at more times of the year so more points on graph/better line (of best fit) on graph; 2. Counted number of individuals in each species so that they could calculate index of diversity; 3. Collected from more sites/more years to increase accuracy of (mean) data;	2 max	Both suggestion and explanation is required for each mark point. 1. The explanation must relate to the graph. 3. For 'accuracy' accept 'representative'.
03.5	1. <i>A. chlorogaster</i> and <i>A. piperi</i> are more closely related (to each other than to <i>P. pruinosa</i>); 2. Because they are in the same genus;	2	1. Must be a comparative statement. 1. Accept <i>A. chlorogaster</i> and <i>A. piperi</i> share a more recent/closer common ancestor (than they do with <i>P. pruinosa</i>); Ignore references to <i>A. chlorogaster</i> and <i>A. piperi</i> not being related to <i>P. pruinosa</i> or not having a common ancestor with <i>P. pruinosa</i>.